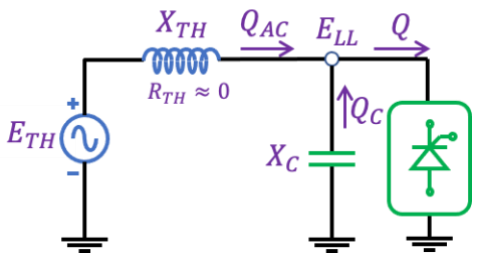
Question 1

a)



$$E_{TH} - E_{LL} \approx Q_{AC}X_{TH} = (Q - Q_C)X_{TH} = \frac{Q - Q_C}{\text{SCL}}$$

- If the AC system voltage (E_{TH}) reduces, the converter terminal voltage (E_{LL}) also reduces resulting in a decrease in the reactive power (Q_C) produced by the local capacitor and filter banks as square of the terminal voltage.

- Considering operation at rated power, the deficit in reactive power ($Q - Q_C$) is transmitted from elsewhere in the AC system which incurs voltage drop over the predominantly reactive transmission lines resulting in further reduction of terminal voltage (E_{LL}) potentially triggering a cascaded problem.

- The extent of voltage drop and hence, potential cascading of the above problem is determined by the Thevenin equivalent impedance (X_{TH}).

- If X_{TH} is high (low) which represents a weak (strong) AC system, the LCC-HVDC would therefore be vulnerable to this cascaded problem.

[1 mark for each point and the diagram – Total 5 marks]

b)

- Direct current control at rectifier (inverter) requires less (high) reactive power at high loading.
- DC fault currents can be automatically limited with current control at the rectifier.
- Voltage control at inverter provides better voltage regulation for loads when in case of HVDC links connecting a generation rich area to a load centre.
- When voltage at the rectifier end reduces below a ceratin level, the rectifier could get stuck at its lower limit of gate delay, where it is unable to control direct current. Inverter takes over current control while the rectifier operates at minimum gate delay.

[1 mark for each point – Total 4 marks]

c)

i) AC-side voltage reduction at rectifier end

- Rectifier end direct voltage and hence the direct current would reduce

Model Answers

- Rectifier controls the current by reducing the gate delay angle α. Once the gate delay angle is stuck at its minimum limit, the inverter takes over current control
- Inverter controls the current by reducing its direct voltage by increasing the extinction advance angle γ. Increase in γ lowers the inverter power factor and results in higher reactive power consumption at the inverter (i.e. opposite) end

[1 mark for each point – Total 3 marks]

ii) AC-side voltage reduction at inverter end

- With constant extinction advance angle, the inverter end direct voltage would reduce and the direct current would increase
- Rectifier controls the direct current by increasing the gate delay angle α
- Increase in α lowers the rectifier power factor and results in higher reactive power consumption at the rectifier (i.e. opposite) end

[1 mark for each point – Total 3 marks]

d) With AC line in operation

i) With AC line

The equivalent reactance at the connection point of the two isolated AC systems

$$x_1 = \frac{V^2}{S_1} = \frac{275^2}{15 \times 10^3} = 5.0417\ \Omega,\ \ x_2 = \frac{V^2}{S_2} = \frac{275^2}{12 \times 10^3} = 6.3021\ \Omega$$

[1 mark]

The equivalent reactance at the connection point of the AC systems with the AC line i.e., $x_l = 5\ \Omega$

$$x_1' = x_1 \parallel (x_l + x_2) = 3.4864\ \Omega$$

$$x_2' = x_2 \parallel (x_l + x_1) = 3.8720\ \Omega$$

[2 marks]

Modified short circuit levels

$$S_1' = \frac{V^2}{x_1'} = 21.69\ \text{GVA},\ S_2' = \frac{V^2}{x_2'} = 19.53\ \text{GVA}$$

[1 mark]

Model Answers

Maximum total rated HVDC capacity at the connection point with system #1

$$P'_{DC1} \leq \frac{S'_1}{m + \mathrm{ESCR}_{\min}} = 5.9428 \text{ GW}$$

[1 mark]

Maximum total rated HVDC capacity at the connection point with system #2

$$P'_{DC2} \leq \frac{S'_2}{m + \mathrm{ESCR}_{\min}} = 5.351 \text{ GW}$$

[1 mark]

Maximum rated capacity of the HVDC link

$$P'_{DC-max} = \min(P'_{DC1}, P'_{DC2}) = \mathbf{5.351\ GW}$$

[1 mark]

[Total - 7 marks]

ii) Without AC line line

When the AC line is out of operation, the system short circuit levels are the same as original values

Maximum total rated HVDC capacity at the connection point with system #1

$$P_{DC1} \leq \frac{S_1}{m + \mathrm{ESCR}_{\min}} = 4.1096 \text{ GW}$$

[1 mark]

Maximum total rated HVDC capacity at the connection point with system #2

$$P_{DC2} \leq \frac{S_2}{m + \mathrm{ESCR}_{\min}} = 3.2877 \text{ GW}$$

[1 mark]

Maximum power transfer through the HVDC link

$$P_{DC-max} = \min(P_{DC1}, P_{DC2}) = \mathbf{3.2877\ GW}$$

[1 mark]

[Total - 3 marks]

Model Answers

Question 2

a) Operating condition #1

No of 6-pulse bridge per pole $B = 2$

Rated current

$$I_d = \frac{P_d/2}{V_d} = 2 \text{ kA}$$

[1 mark]

Inverter end maintains the rated voltage $V_{di} = 500$ kV while operating at the minimum extinction angle by adjusting the effective turns ratio of the converter transformer

[1 mark]

$$V_{d0i} = \frac{V_{di} + BR_cI_d}{\cos(\gamma_{min})} = \frac{3\sqrt{2}}{\pi} BT_iE_{ac} \Rightarrow T_i = \frac{(V_{di} + BR_cI_d)\pi}{3\sqrt{2}BE_{ac}\cos(\gamma_{min})}$$

[1 mark]

With $E_{ac} = E_{ac-rated} = 400$ kV

$$T_i = \frac{(V_{di} + BR_cI_d)\pi}{3\sqrt{2}BE_{ac-rated}\cos(\gamma_{min})} = \mathbf{0.5083}$$

[1 mark]

$$V_{d0i} = \frac{3\sqrt{2}}{\pi} BT_iE_{ac-rated} = 549.12 \text{ kV}$$

Power factor of inverter

$$\cos\phi_i = \frac{V_{di}}{V_{d0i}} = 0.9106$$

[1 mark]

Reactive power consumption of inverter

$Q_i = 2 \times V_{di} \times I_d \times \cos\phi_i = \mathbf{907.99\ MVAr}$

[1 mark]

Direct voltage at rectifier end

$V_{dr} = V_{di} + R_LI_d = 508$ kV

[1 mark]

Model Answers

Rectifier end maintains the rated current while operating at the nominal firing angle by adjusting the effective turns ratio of the converter transformer

$$T_r = \frac{(V_{dr} + BR_c I_d)\pi}{3\sqrt{2}BE_{ac-rated}\cos(\alpha)} = \mathbf{0.5072}$$

[1 mark]

$$V_{d0r} = \frac{3\sqrt{2}}{\pi} BT_r E_{ac} = 547.94 \text{ kV}$$

Power factor of inverter

$$\cos\phi_r = \frac{V_{dr}}{V_{d0r}} = 0.9271$$

[1 mark]

Reactive power consumption of rectifier

$$Q_r = 2 \times V_{dr} \times I_d \times \cos\phi_r = \mathbf{821.48\ MVAr}$$

[1 mark]

[Total - 10 marks]

b) Operating condition #2

Here the inverter continues to maintain the rated voltage while the rectifier maintains the rated current while operating with the nominal gate delay angle. The rectifier end AC system voltage and direct voltage stays the same. So both reactive power consumption and turns ratio of the converter transformer on the rectifier end remains the same as under normal condition.

[2 marks]

Reduced AC voltage at the inverter end

$$E_{ac-i}{}' = 0.92 \times 400 \text{ kV} = 368 \text{ kV}$$

[1 mark]

$$T_i{}' = \frac{(V_{di} + BR_c I_d)\pi}{3\sqrt{2}BE_{ac-i}{}'\cos(\gamma_{min})} = \mathbf{0.5525}$$

[1 mark]

The turns ratio basically compensates for the reduction in AC system voltage keeping V_{d0i} same as in normal condition. So the reactive power consumption of the inverter remains the same.

[1 mark]

Model Answers

[Total - 5 marks]

c) Operating condition #3

Reduced AC voltage at the rectifier end

$E_{ac-r}{}' = 0.88 \times 400 \text{ kV} = 352 \text{ kV}$

Revised gate delay angle needed to maintain the direct current

$$\cos(\alpha) = \frac{(V_{dr} + BR_c I_d)\pi}{3\sqrt{2}BT_r E_{ac-r}{}'} > 1$$

[1 mark]

There is no gate delay angle that allows the rectifier to maintain the direct current. So the inverter takes over current control with a modified current reference.

$$I_d{}' = I_d\left(1 - \frac{I_m}{100}\right) = 1.7 \text{ kA}$$

[1 mark]

At the rectifier end, the gate delay angle is stuck at the minimum value - so the effective turns ratio is adjusted to maintain the rated direct voltage.

[1 mark]

$$T_r{}' = \frac{(V_{d-rated} + BR_c I_d{}')\pi}{3\sqrt{2}BE_{ac-r}{}'\cos(\alpha_{min})} = \mathbf{0.5415}$$

[1 mark]

$$V_{d0r}{}' = \frac{3\sqrt{2}}{\pi}BT_r{}'E_{ac-r}{}' = 514.85 \text{ kV}$$

Power factor of rectifier

$V_{dr}{}' = V_{d-rated} = 500 \text{ kV}$

$$\cos\phi_r{}' = \frac{V_{dr}{}'}{V_{d0r}{}'} = 0.9711$$

[1 mark]

Reactive power consumption of rectifier

$Q_r{}' = 2 \times V_{dr}{}' \times I_d{}' \times \cos\phi_r{}' = \mathbf{417.45 \text{ MVAr}}$

[1 mark]

Model Answers

Direct voltage at the inverter end

$V_{di}' = V_{dr}' - R_L I_d' = 493.20$ kV

[1 mark]

At the inverter end, the effective turns ratio is adjusted to produce this direct voltage with minimum extinction angle.

$$T_i' = \frac{(V_{di}' + BR_c I_d')\pi}{3\sqrt{2}BE_{ac-rated}\cos(\gamma_{min})} = \mathbf{0.4992}$$

[1 mark]

$$V_{doi}' = \frac{3\sqrt{2}}{\pi} BT_i' E_{ac-rated} = 539.3253 \text{ kV}$$

Power factor of rectifier

$$\cos\phi_i' = \frac{V_{di}'}{V_{doi}'} = 0.9145$$

[1 mark]

Reactive power consumption of rectifier

$Q_i' = 2 \times V_{di}' \times I_d' \times \cos\phi_i' = \mathbf{741.99\ MVAr}$

[1 mark]

[Total - 10 marks]

Model Answers

Question 3

a)

(i)
With no current in the circuit, there are no AC or DC voltage drops across the resistance or the inductance of the circuit, therefore: $V_o^{DC} = V_{DC}$ =700 V and $V_o^{AC} = -V_{AC}$ =400 with angle 180º.
[2 mark]

(ii)
The output voltage of the submodule is a time-varying signal resulting from the summation of the DC voltage and the AC voltage. It's highest value is at the peak of the sine, this gives: $V_o^{peak} = V_o^{DC} + \sqrt{2}\|V_o^{AC}\|$ =1.27 kV. The submodule in the diagram is a half-bridge submodule, which means that it can generate an output voltage V_o in the range between 0 a V_{SM}. Therefore, the minimum submodule voltage required is V_{SM}^{min} =1.27 kV.
[2 mark]

(iii)
Half-bridge submodules can only apply positive voltage. The minimum voltage the converter must apply in order to remain at zero current is $V_o^{peak-low} = V_o^{DC} - \sqrt{2}\|V_o^{AC}\|$. If $V_o^{DC}$was too low, the required $V_o^{peak-low}$to keep the current controlled at zero would become negative, which is impossible to generate with a half-bridge submodule. Consequently, the minimum DC voltage of operation is $V_o^{DCmin} = \sqrt{2}\|V_o^{AC}\|$ =567 V (that's 81% of the nominal DC voltage).
[2 mark]

b) [Bookwork] When comparing two submodules of identical DC-side voltage rating but half-bridge against full-bridge, the most apparent advantage of the full-bridge is that it can generate negative voltages, which allows for modular multilevel converter designs that can withstand DC short-circuits without losing control of the fault current. Additionally, the submodule allows three possible output voltage levels (ie V_{SM}, 0 and $-V_{SM}$). On the negative side, a half-bridge submodule makes the current flow through one of the semiconductor devices (IGBT or diode), whereas the full-bridge has two semiconductor devices in series conducting the current. This causes the full-bridge submodule to have approximately twice as many conduction losses, which is a key parameters for very high power designs.
[3 mark]

c)

(i)
Zero net power balance means that the power the submodule exchanges with the rest of the circuit averages to zero, ie the internal energy of the submodule doesn't build up or deplete over time. This implies that the power exchanged using AC voltage and current counteracts the power exchanges using DC voltage and current, ie: $\mathbb{R}e\{V_o^{AC}(I_o^{AC})^*\} + V_o^{DC}I_o^{DC} = 0$.
[2 mark]

(ii)
The power reference gives the amount of power exchanged with the AC grid; therefore, the AC current can be calculated as:$I_{AC} = \frac{P}{\|V_{AC}\|} - j\frac{Q}{\|V_{AC}\|}$ =500 A (taking the AC grid voltage as the angle reference).

Model Answers

Next we can find what AC voltage the converter would need to apply in order for the current to match this reference value. This gives: $V_o^{AC} = -V_{AC} - (R + jX)\, I_{AC}$ =420 V with angle -169º.
Once we know the AC voltage of the stack and the AC current, we can calculate the power the submodule exchanges through AC, this gives: $P_o^{AC} = \mathbb{R}e\{V_o^{AC}(I_o^{AC})^*\}$ =-206 kW. The difference of 6 kW compared with the power of the grid is due to the conduction losses of the bridge and the output filter.
The power exchanged using DC must match this value with opposite sign, this gives: P_o^{DC} =206 kW.
[5 mark]

d) Ccc

(i)
We can calculate the power exchanged with the AC grid as: $S_{AC} \triangleq P_{AC} + jQ_{AC} = VI^*$ =400V 100A, ie P_{AC} =40 kW (receiving), Q_{AC} =0. And the power of the DC grid as $P_{DC} = V_{DC}I_{DC}$ =700V 200 A=140 kW (supplying). The large difference between P_{AC} and P_{DC} indicate that the internal energy of the converter is building-up.
[1 mark]

(ii)
The equation that relates AC voltages and currents is: $V_o^{AC} = -V_{AC} - (R + jX)\, I_{AC}$ =403 V at an angle of -178º. As for the DC components, if we neglect the resistance of the circuit we get $V_o^{DC} = V_{DC}$ = 700 V. We can test if this induces a big error, the voltage drop caused by the current is $V_R^{DC} = R\, I_{DC}$ = 24 mΩ 200A = 4.8 V, which is about 0.7% (ie very small).
[2 mark]

(iii)
We can calculate the apparent power of the submodule as $S_o \triangleq P_o^{AC} + jQ_o = V_o^{AC}\{I_{AC}\}^*$ =-40.2 kW -1.6 kVAr. And the DC part is $P_o^{DC} \approx P_{DC}$ = 140 kW. The net power is $P_{net} = P_o^{AC} + P_o^{DC}$ =99.8 kW, ie the submodule's internal energy is raising at this rate. As for the reactive power, the 1.6 kVAr supplied to the circuit feeds the reactive power of the reactor of the filter.
[3 mark]

e)

(i)

The transfer function between V_o and I can be obtained from the circuit equations, which are linear:

$$-V_o(t) - V_{AC}(t) + V_{DC}(t) = R\, I(t) + L\frac{d}{dt}I(t)$$

Transforming by Laplace this gives

$$-V_o(s) - V_{AC}(s) + V_{DC}(s) = R\, I(s) + LsI(s)$$

Making all other inputs zero and solving for $I(s)$ we can find the transfer function $G(s)$ defined as:

$$G(s) \triangleq \frac{I(s)}{V_o(s)} = \frac{-1}{Ls + R}$$

[1 mark]

(ii)

Model Answers

The closed-loop controller is to obtain the value of V_o to be applied from a PI fed with the error between the current reference and the actual current, ie $V_o(s) := K(s)(I^*(s) - I(s))$ with $K(s) \triangleq \frac{K_P s + K_I}{s}$.

We can multiply this by $G(s)$ and solve for the closed loop transfer functione defined as $T(s) \triangleq \frac{I(s)}{I^*(s)}$.
This gives

$$I(s) = G(s)V_o(s) = G(s)K(s)(I^*(s) - I(s))$$

which leads to

$$T(s) = \frac{I(s)}{I^*(s)} = \frac{G(s)K(s)}{1 + G(s)K(s)} = \frac{\frac{-1}{Ls+R}\frac{K_P s + K_I}{s}}{1 + \frac{-1}{Ls+R}\frac{K_P s + K_I}{s}} = \frac{-K_P s - K_I}{Ls^2 + (R - K_P)s - K_I}$$

The exercise gives a specific choice for the PI controller that cancels out some of the terms in this derivation. Students may use this for their advantage:

$$T(s) = \frac{I(s)}{I^*(s)} = \frac{G(s)K(s)}{1 + G(s)K(s)} = \frac{\frac{-1}{Ls+R}\frac{K_P s + K_I}{s}}{1 + \frac{-1}{Ls+R}\frac{K_P s + K_I}{s}} = \frac{\frac{-1}{Ls+R}\frac{-Ls - R}{\tau^* s}}{1 + \frac{-1}{Ls+R}\frac{-Ls - R}{\tau^* s}} = \frac{1}{\tau^* s + 1}$$

[2 mark]

(iii)

Next we are asked to find the frequency response at 50 Hz, which we can obtain by replacing s by $j\omega$ with $\omega := 2\pi 50$ and check for the amplitude and the phase of the resulting complex number.

If students skipped tasks i and ii before, they could take the closed-loop transfer function that is hinted in the introduction of question 3. This would be taken as valid with no marks subtracted.

$T(j\ 2\ \pi\ 50)$ =0.53 – j0.5

Taking the norm and the angle, we find that the current would have an amplitude of 0.73 A at a phase delay of -43º.
[3 mark]

(iv)

This question requires us to go back to the original voltage and current equation where the other voltages are considered:

$$-V_o(s) - V_{AC}(s) + V_{DC}(s) = R\ I(s) + LsI(s)$$

Solving for $I(s)$ we get:

$$I(s) = \underbrace{\frac{-1}{Ls+R}}_{G(s)} V_o(s) + \underbrace{\frac{-1}{Ls+R}}_{G_{d1}(s)} V_{AC}(s)$$

where we can skip V_{DC} given that we only care about the closed-loop control of the current and we'll consider the reference $I^*(s)$ and $V_{DC}(s)$ to be zero in order to study the effect of $V_{AC}(s)$. Introducing the controller we get:

$$I(s) = G(s)K(s)(\underbrace{I^*(s)}_{0} - I(s)) + G_{d1}(s)V_{AC}(s)$$

Model Answers

This gives:

$$I(s) = \frac{G_{d1}(s)}{1 + G(s)K(s)} V_{AC}(s)$$

Subtituting, we get

$$I(s) = \frac{\frac{-1}{Ls + R}}{1 + \frac{-1}{Ls + R}\frac{K_P s + K_I}{s}} = \frac{-s}{Ls^2 + (R - K_P)s - K_I}$$

[2 mark]

Model Answers